

MATHEMATICAL OPTIMIZATION: LAB

ABSTRACT. Tasks for the LAB

1. LAB 1

1. Install Julia + Jupyter Lab + JuMP solver framework + SCS solver and work through the linear programming Jump tutorial https://jump.dev/JuMP.jl/stable/tutorials/getting_started/getting_started_with_JuMP/

Apart from Julia itself (and perhaps jupyter lab) you should be able to install most of this through the Julia package manager.

2. LAB 2

2. Check *strong duality* and *complementary slackness* for numerical solutions.
3. Dualize an infeasible LP and obtain infeasibility certificate

3. LAB 3

4. Implement LP relaxations for maximum perfect matching and vertex cover.
5. Implement the relaxations FSTAB, QSTAB for the stable set polytope.

4. LAB 4

6. Implement LP's for FSTAB and QSTAB.

5. LAB 5

7. Let $P = \text{conv}\{(0, 0), (2, 0), (0, 1), (2/3, 2/3)\}$. Describe the Anti-blocker P^* .
8. Recall:

Theorem 1 (Chvatal for positive orthant). *Given*

$$(1) \quad \min\{c^T x : Ax \leq b, x \in \mathbb{Z}_+^n\}$$

where $A \in \mathbb{Z}^{m \times n}, b \in \mathbb{Z}^m, c \in \mathbb{R}$; the inequality

$$(2) \quad \lfloor \lambda^T A \rfloor x \leq \lfloor \lambda^T b \rfloor$$

is valid for P_I for all $\lambda \in \mathbb{R}_{\geq 0}^m$.

Task: Example from <https://coral.ise.lehigh.edu/ted/files/ie418/lectures/Lecture13.pdf>
Make numerical iterative Gomory-Chvatal cuts for

$$(3) \quad \begin{array}{ll} \max & 2x_1 + 5x_2 \\ \text{s.t.} & 4x_1 + x_2 \leq 28 \\ & x_1 + 4x_2 \leq 27 \\ & x_1 - x_2 \leq 1 \end{array}$$

$$(4) \quad x_1, x_2 \geq 0$$

(Optimal basic feasible solution is: $x_1^* = 17/3, x_2^* = 16/3$ with $\alpha^* = 38$. Optimal IP solution is $\alpha_{IP}^* = 37.5$)

Can you find a numerical Gomory-Chvatal Closure P' ?

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6. LAB 6

9. Eigenvalues, trace, scalar product, matrix product, Cholesky decomposition

7. LAB 7

10. Study the Max Cut Note, and try to understand each line in the derivation (pen + paper, make examples). Implement the SDP.

8. LAB 8

11. Study the SOS Note, and try to understand each line in the derivation (pen + paper, make examples). Then find a decomposition of $5x^4 - 4x^3 - x^2 + 2x + 2$ as a sum of squares polynomial: first by hand and then by computer.

9. LAB 9

12. Compute $\vartheta(G^{\boxtimes 2})$ where G is the pentagon and $\boxtimes$ the strong graph product.